

Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be iid r.v.s with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$. Let $S_n = X_1 + X_2 + \dots + X_n$ for $n \geq 1$.

Then, the standardized sum converges in distribution to a standard normal:

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \xrightarrow{d} N(0, 1)$$

- **Statistic of interest:** $\frac{S_n - n\mu}{\sqrt{n}\sigma}$
- **Centering:** $S_n - n\mu$
- **Scaling:** $\sqrt{n}\sigma$

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Note:

1. $n\mu = \mathbb{E}[S_n]$
2. $\sqrt{n}\sigma = \text{S.D.}(S_n)$

In other words, CLT states:

$$\frac{S_n - \mathbb{E}[S_n]}{\text{S.D.}(S_n)} \xrightarrow{d} N(0, 1)$$

This **Standardised Statistic of interest** can also be written in terms of the sample mean, $\bar{X}_n$:

$$\frac{S_n - \mathbb{E}[S_n]}{\text{S.D.}(S_n)} = \frac{S_n - n\mu}{\sqrt{n}\sigma} = \frac{n(\bar{X}_n - \mu)}{\sqrt{n}\sigma} = \sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right)$$

Recall: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i = \frac{S_n}{n} \iff n\bar{X}_n = S_n$.

Thus, the CLT also states:

$$\sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right) \xrightarrow{d} N(0, 1)$$

Comparison with Laws of Large Numbers

- **WLLN:** $\bar{X}_n \xrightarrow{P} \mu$
- **SLLN:** $\bar{X}_n \xrightarrow{a.s.} \mu$

Thus, $(\bar{X}_n - \mu) \rightarrow 0$ (in probability and almost surely). From statistics course we know $\bar{X}_n$ is an unbiased estimator of μ .

The CLT describes the rate of this convergence. The error term, $(\bar{X}_n - \mu)$, when scaled by $\frac{\sqrt{n}}{\sigma}$, has a non-trivial limit distribution which is $N(0, 1)$:

$$\frac{\sqrt{n}}{\sigma}(\bar{X}_n - \mu) \xrightarrow{d} N(0, 1)$$

So we can calculate the probability of the error being within a certain tolerance t :

$$P(|\bar{X}_n - \mu| \leq t) = P\left(\left|\frac{\sqrt{n}}{\sigma}(\bar{X}_n - \mu)\right| \leq \frac{\sqrt{nt}}{\sigma}\right)$$

Let $Z \sim N(0, 1)$, then:

$$P(|\bar{X}_n - \mu| \leq t) \approx P\left(|Z| \leq \frac{\sqrt{nt}}{\sigma}\right) = 2\Phi\left(\frac{\sqrt{nt}}{\sigma}\right) - 1$$

where Φ is the CDF of the $N(0, 1)$ distribution. $\sqrt{n}$ is the scale of the error. But here σ may not be known and also we are not sure about how good is this approximation?

Random Walk (RW)

$S_n = X_1 + X_2 + \dots + X_n$ is called an iid Random Walk (RW).

Simple Symmetric Random Walk (SSRW)

Let $X_1, \dots, X_n$ be iid such that $P(X_i = 1) = P(X_i = -1) = \frac{1}{2}$.

- **Mean:** $\mu = \mathbb{E}[X_i] = (1)(\frac{1}{2}) + (-1)(\frac{1}{2}) = 0$
- **Variance:** $\sigma^2 = \mathbb{E}[X_i^2] - \mu^2 = ((1)^2(\frac{1}{2}) + (-1)^2(\frac{1}{2})) - 0^2 = 1$

For the SSRW:

- **WLLN:** $\frac{S_n}{n} \xrightarrow{P} 0$
- **SLLN:** $\frac{S_n}{n} \xrightarrow{a.s.} 0$
- **CLT:** $\frac{S_n - n(0)}{\sqrt{n(1)}} = \frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0, 1)$

This implies that for a large c (e.g., $c = 10$):

$$P\left(\left|\frac{S_n}{\sqrt{n}}\right| \geq c\right) \approx 2(1 - \Phi(c)) > 0$$

This suggests that $|S_n| \approx c\sqrt{n}$. The magnitude of the walk grows at the scale of $\sqrt{n}$.

Example: Product of RVs

Let $T_n = \prod_{i=1}^n Y_i$, where $Y_i > 0$ are iid. We can apply the CLT to $\log T_n$:

$$\log T_n = \sum_{i=1}^n \log Y_i$$

Let $X_i = \log Y_i$. Let $\mu_X = \mathbb{E}[X_i]$ and $\sigma_X^2 = \text{Var}(X_i)$. By CLT:

$$\frac{\log T_n - n\mu_X}{\sqrt{n}\sigma_X} \xrightarrow{d} N(0, 1)$$

where $\mathbb{E}[\log T_n] = n\mathbb{E}[\log Y_i] = n\mu_X$ and $\text{S.D.}(\log T_n) = \sqrt{n}\text{S.D.}(\log Y_i) = \sqrt{n}\sigma_X$.

Confidence Intervals for μ

Let $X_1, X_2, \dots, X_n$ be iid with mean μ and variance σ^2 . We want to find an interval (L_n, U_n) such that $P(\mu \in (L_n, U_n)) \approx 0.95$. $\bar{X}_n$ is an unbiased estimator of μ .

Case 1: σ^2 is known

From the CLT, we know:

$$\sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right) \xrightarrow{d} Z \sim N(0, 1)$$

We find z such that $P(-z \leq Z \leq z) = 0.95$. This gives $z \approx 1.96$ (since $\Phi(1.96) \approx 0.975$).

$$P \left(-1.96 \leq \sqrt{n} \frac{(\bar{X}_n - \mu)}{\sigma} \leq 1.96 \right) \approx 0.95$$

Rearranging the inequality to isolate μ :

$$\iff P \left(\bar{X}_n - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + 1.96 \frac{\sigma}{\sqrt{n}} \right) \approx 0.95$$

So, the 95% confidence interval is:

$$\left[\bar{X}_n - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X}_n + 1.96 \frac{\sigma}{\sqrt{n}} \right]$$

Issue: In practice, σ^2 is almost always unknown.

Case 2: σ^2 is unknown

We must estimate σ^2 from the data.

- **Natural estimate (MLE):** $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$
- **Unbiased estimate (Sample Variance):** $\hat{\sigma}_n^2 = S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$

What if we just plug in $\hat{\sigma}_n$ for σ in the interval formula?

$$\left[\bar{X}_n - 1.96 \frac{\hat{\sigma}_n}{\sqrt{n}}, \bar{X}_n + 1.96 \frac{\hat{\sigma}_n}{\sqrt{n}} \right]$$

Does this interval have 95% confidence? i.e., is it true that $P\left(\left|\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\hat{\sigma}_n}\right)\right| \leq 1.96\right) \approx 0.95$?

This requires the statistic $\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\hat{\sigma}_n}\right)$ to have an approximate $N(0, 1)$ distribution. If n is not big enough then the distribution is not $N(0, 1)$ and our derived result won't hold true. If n is large enough we prove it by Slutsky's Theorem.

Theorem: Slutsky's Theorem

Suppose $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} c$ (where c is a constant). Then:

1. $X_n + Y_n \xrightarrow{d} X + c$
2. $X_n Y_n \xrightarrow{d} cX$
3. $X_n / Y_n \xrightarrow{d} X/c$, if $c \neq 0$

Application to the Confidence Interval:

1. **From CLT:** Let $X_n = \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right)$. We know $X_n \xrightarrow{d} N(0, 1)$.
2. **From WLLN:** The sample variance $\hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.
We can write $\hat{\sigma}_n^2 = \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n X_i^2 - (\bar{X}_n)^2\right)$.
By WLLN, $\bar{X}_n \xrightarrow{P} \mu$ and $\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{P} \mathbb{E}[X_i^2]$.
As $n \rightarrow \infty$, $\frac{n}{n-1} \rightarrow 1$. Therefore, $\hat{\sigma}_n^2 \xrightarrow{P} 1(\mathbb{E}[X_i^2] - \mu^2) = \text{Var}(X_1) = \sigma^2$.
3. **By Continuous Mapping Theorem:** Since $g(x) = \sqrt{x}$ is continuous, $\hat{\sigma}_n = \sqrt{\hat{\sigma}_n^2} \xrightarrow{P} \sqrt{\sigma^2} = \sigma$.

Now, let $X_n = \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right)$ and $Y_n = \frac{\hat{\sigma}_n}{\sigma}$. We have $X_n \xrightarrow{d} N(0, 1)$ and $Y_n \xrightarrow{P} \frac{\sigma}{\sigma} = 1$. By Slutsky's Theorem (part 3):

$$\frac{X_n}{Y_n} = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{\hat{\sigma}_n/\sigma} = \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\hat{\sigma}_n}\right) \xrightarrow{d} \frac{N(0, 1)}{1} = N(0, 1)$$

Thus, the 95% confidence interval using $\hat{\sigma}_n$ is asymptotically valid.

Thus 1.96 will work :)

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